Charbel Barrak

Computer Science

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EDUCATION

Bachelor's degree in computer science
Lebanese University, 2023 - 2026

Masters in computer science (In Progress)
Lebanese University, 2026 - Present

B2 – French Language Certificate
French Embassy - Beirut, Jan 2022

Links

Github ,LinkedIn

Technical Skills

- Object-Oriented Programing (OOP)
- Front-End Development (React.js, JavaScript, CSS, BootStrap)
- Back-End Development (Node.js , FastAPI,RESTful APIs)
- Mobile Application Development (Java ,Swift)
- AI & Machine Learning (Prompt Engineering, LLMs, NLP)
- DataBase Management (MySQL, MongoDB)
- Version Control (Git, Github, Postman)

Projects

1) AI-Based Stock Market Prediction Web Application

- Developed an AI-powered stock prediction platform to analyze historical market data and generate trend forecasts.

- Implemented machine learning models for financial data analysis and predictive insights.
- Built data processing pipelines and visualization features to present market trends and predictions.
- Designed a React-based user interface for displaying stock insights and model results

Technologies used: Python, TensorFlow, Machine Learning , Pandas

Repository-Link: https://github.com/charbell2005/FYP-AIStockPredictionSystem

2) Full-Stack Inventory Management Web Application

- Designed and developed a full-stack application for inventory, supplier, and order management.
- Created scalable REST APIs using ASP.NET Core and integrated them with a React-based frontend.
- Implemented CRUD operations, analytics dashboards, and database management using LiteDB.
- Improved application reliability through error handling and API fallback strategies.

Technologies used: React, Vite, ASP.NET Core, C#, LiteDB, REST APIs

Repository-Link: https://github.com/charbell2005/SmartStockPro

3) Pathfinding Algorithm Visualizer

- Developed an interactive maze pathfinding visualizer comparing BFS, DFS, and A* algorithms side by side.
- Implemented real-time animated exploration with live node tracking and queue/stack state visualization.
- Built using Python and Pygame with automated best-path selection and comparison output.

Technologies used: Python, Pygame, Matpolib, Algorithms

Repository-Link: https://github.com/charbell2005/AI-MazeProject

4) 3D Graphics Application

- Built a real-time 3D scene using C++ and OpenGL with object transformations,

- rendering pipelines, and lighting
- Applied core computer graphics principles including projection, shading, and rasterization
- Technologies used: C++, OpenGL

Professional Experience

Freelance Web Developer, Self-Employed
Jun 2025 – Present

- Developed responsive websites and Full-Stack web applications for clients based on business requirements.
- Designed and implemented frontend interfaces using modern web technologies.
- Integrated APIs, databases, and backend services to deliver functional web solutions.
- Collaborated with clients to gather requirements, provide updates, and deliver projects on time.

Languages

- French (Fluent)
- English (Fluent)